

ALEX ● KASKASOLI

README

Security engineer with an offensive security background. My last role was at Google DeepMind, where among other things I threat modelled the production deployment of ML solutions and lead the initial framework for cybersecurity safety evaluations, including running agents and vulnerable-by-design systems in isolated k8s environments, navigating Google's strict security policies.

In my previous roles as a Security Engineer and Head of Security and DevOps, I drove the design of AWS organizations from scratch, hardened k8s clusters and cloud environments, pushed GitOps practices across the engineering pool, and set up logging and monitoring stacks. I also worked alongside ISO 27001 consultants, translating compliance requirements into infrastructure-as-code policies.

Back in my pentesting days, I conducted security assessments on a range of tech stacks and participated in "red team" engagements against large organizations, often exploiting then-nascent CI/CD and DevOps tooling for lateral movement and privilege escalation.

BLOG

My blog <https://alex.kaskaso.li/> showcases some personal work, conference talks and open-source projects. This includes a critical vulnerability in Kubernetes' Minikube, some thoughts on CI/CD security that were cited on Google's k8s podcast, a custom k8s controller, a framework for conducting DNS rebinding attacks as well as an interesting attack on AWS cloud environments using headless browsers (nominated for PortSwigger's "best hacks of 2018").

SKILLS

Cloud & IAM

Designed multi-account **AWS** orgs from scratch, including org-wide CloudTrail/S3/CloudWatch log centralization with **SCP** guardrails. Built bespoke **IAM** tooling to balance security and developer velocity.

CI/CD

Built **GitOops**, a tool that briefly trended on GitHub, for attacking and defending CI/CD pipelines. Early **GitOps** adopter for infra-as-code, wrote and spoke on CI/CD security and pipeline hardening.

Kubernetes

Researched and published **k8s** privilege-escalation paths (including a CVE for RCE/VM escape in minikube). Ran and hardened production clusters, and wrote custom k8s controllers for bespoke needs.

Email & Comms Security

Built **MailspooF**, an open-source SPF/DMARC scanner for email spoofing exposure, and wrote on the topic. Practical experience hardening collaboration and comms security (emails, Slack).

Coding

Ship production tooling in **Python** and **Go**. Comfortable in **JavaScript**, some **C** and **Assembly** for offensive tooling and basic memory corruption research.

ML / AI Security

Threat modelled ML deployments and watermarking robustness at Google DeepMind. A Statistics & Data Science MicroMasters from MIT allows me to speak comfortably with researchers.

EXPERIENCE	Google DeepMind <i>Security SWE, threat modelling ML deployments and moonlighting on some adversarial testing of image and text watermarks. Built the initial framework for cybersecurity safety evaluations.</i>	3 years <i>London, UK</i>
	Kaluza <i>Tech Lead Security Engineer at Kaluza, working on custom k8s secrets management for the new platform, centralized infrastructure logging for a large AWS org, automating security controls, identifying attack paths across the cloud environments and advising engineers on closing gaps.</i>	2 years <i>London, UK</i>
	TrueLayer <i>Senior Cloud Security Engineer promoted to Head of DevOps and Security, working closely with engineering teams and upper management to scale infrastructure, enhance DevOps practices, mitigate security risks and meet regulatory compliance.</i>	1.5 years <i>London, UK</i>
	MWR InfoSecurity <i>Security Consultant at MWR InfoSecurity, working on infrastructure and application security and simulated attacks ("red team"). Worked with clients across a range of industries from finance to critical infrastructure.</i>	4 years <i>London, UK</i>
CERTIFICATES	OSCP, OSCE, CRT, CCT MicroMasters Statistics and Data Science (MITx) BSc Ethical Hacking & Countermeasures (University of Abertay)	
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